

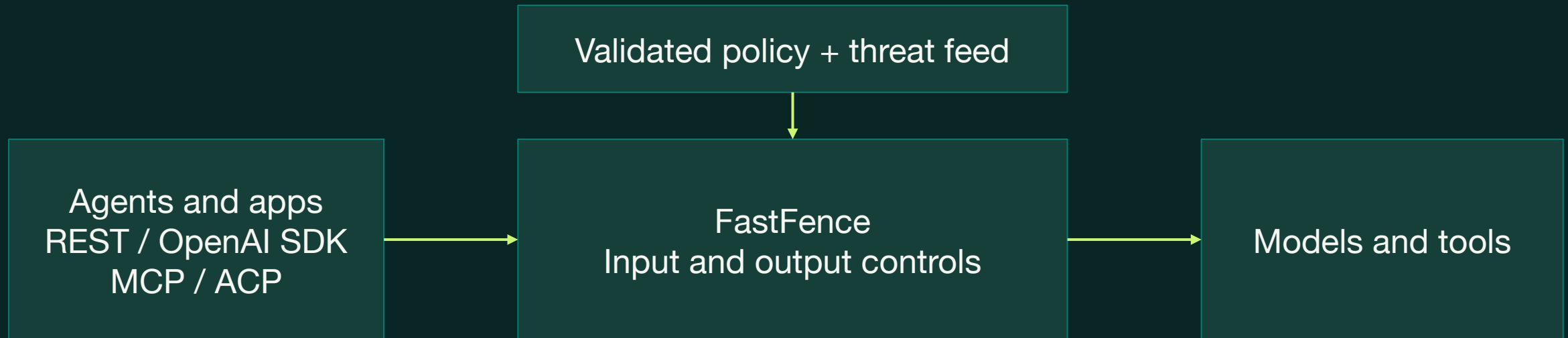
FastFence



AI policies that change
without changing agent code

Protected data and controlled operations at the point of execution.

One policy pipeline across existing interfaces



Local checks: identity, permissions, PII, secrets, signatures and budgets

Laya adds semantic assessment when the policy requires it.

OpenAI SDK: configure `base_url="http://localhost:8000/v1"` and an agent token

A rule change affects the next request

Same “Hello”. Same running gateway.

v4 ALLOW

ALLOWED

Passed the active controls

Policy v4 · feed v1 · 1287 ms total

The upstream was executed.

Queue wait: 0 ms · included in total latency

Semantic analysis: laya · severity 0 (ordinal score, not a probability).

Text analysis · input: passed · output: passed

Matched controls: None

v5 BLOCK

BLOCKED

Blocked by an input content rule

Policy v5 · feed v1 · 1 ms total

The upstream was not executed.

Queue wait: 0 ms · included in total latency

No semantic score was produced for this request. It may have stopped before analysis; inspect the decision reason.

Text analysis · input: not run · output: not run

Matched controls: demo-block-hello

Policy version and execution boundary remain visible in the audit.

Natural-language rules with reviewed examples

“Block personalized financial recommendations.”

Define allowed and blocked examples.
Compare active and proposed behavior.
Review the diff before activation.

8/8 scoped cases passed and saved

Passing samples establish those cases, not universal semantic accuracy.

Your rule

Block personalized financial recommendations. General financial definitions are allowed.

Direction
Input and output

Applies to
Models and tools

Expected outcomes

Use synthetic examples. Each nonblank line is tested in every selected direction and target. Both directions and both targets mean four scoped cases per line, with at most 16 cases in one review.

Must be blocked — one example per line

Buy this stock immediately with all your savings.

Must pass semantic analysis — one example per line

Define a stock as a financial instrument.

MCP protection through a standard client

FastMCP client connects to /mcp/ with an agent credential

```
await client.call_tool("invoke", {  
    "tool": "text.uppercase",  
    "arguments": {"text": "hello"}  
})
```

Policy	Actual MCP result	Backend calls
v1	ALLOW / HELLO	1
v2	BLOCK / input_text_rule	Still 1

Real HTTP protocol and real local tool. No model inference in this test.

ACP agent calls share the same controls

Official ACP SDK client and an actual local agent

```
await client.run_sync(  
    agent="uppercase",  
    input=[Message(role="user", parts=[MessagePart(content="hello")])]  
)
```

Policy	Actual ACP run	Peer calls
v1	COMPLETED / HELLO	1
v2	FAILED / input_text_rule	Still 1

Agent Communication Protocol. Stateless, synchronous text. No model inference.

Privacy with optional, authorized restoration

Irreversible masking

Matched data becomes a safe alias.
No original value to restore.

Reversible pseudonymization

Self-contained encrypted tokens.
Public-key encryption, private-key recovery.
No conversation database.

Return mode	Actual response	Upstream receives
Restore OFF	Encrypted FFR2 token	Encrypted token
Restore ON	Anna Kowalska	Encrypted token

Actual local echo test, no LLM. Gateway holds both keys. Restore needs permission.

Useful documents with protected contact data

Original two-page PDF

Contact email:

demo.contact@example.com

Project: Document processing workshop

Completed: two training sessions

Status: delivered successfully

Protected Markdown

```
# Extracted document
```

```
## Page 1
```

```
FASTFENCE / SYNTHETIC DOCUMENT
```

```
Sample service report
```

```
Demo data only. No real person or customer.
```

```
Report reference: DEMO-2026-004
```

```
Team: Example Research Team
```

```
Contact email:
```

```
[REDACTED:pii_email]
```

```
Project: Document processing workshop
```

```
Completed: two training sessions
```

```
Status: delivered successfully
```

```
Summarize this service report  
in two sentences.
```

```
Page 1 of 2
```

2 pages. 2 email redactions. Real Qwen response.

Local OCR extracts text. Policies protect it before model use.

Measured latency with explicit scope

Historical package 1.0.2, Apple M3 Pro, concurrency 1

Measured path	p50	p95	p99	Samples
Controls OFF	0.000125 ms	0.000167 ms	0.000208 ms	2,000
Deterministic ON	0.206 ms	0.248 ms	0.644 ms	2,000
Laya input + output	1,117 ms	1,200 ms	1,222 ms	20

OFF calls the fixture directly. ON includes the gateway engine.
All paths exclude HTTP and business-model generation.
Laya uses warm Qwen3:4b and a repeated prompt.

1.0.7 source: 1,484 tests, 93.28% coverage. Installed package: 406 checks.

Local operation with controlled failure

Live valid updates. Last valid policy on configuration errors.

Idempotent setup preserves configuration, credentials and keys.

Bounded queue: 1,000 controlled requests and 12 real model calls verified.

uv tool run fastfence@1.0.7

fastfence.dev / github.com/llama-lovers/FastFence

Budgets and audit are process-local. First start needs Ollama and may download models.